November 14, 2026

Editors, *Journal of Chemical Information and Modeling* (ACS)

Dear Editors,

I am pleased to submit our manuscript, "Three-Dimensional Pocket Geometry Does Not Beat Its Controls: A Measured Test of Structural Grounding for Protein Foundation Model Affinity Prediction on 41,271 ChEMBL Pairs," for consideration as an Article in the *Journal of Chemical Information and Modeling*.

This is the pre-registered sequel to my companion study currently under review at JCIM (Manuscript ID ci-2026-04206a), which measured that 1D sequence-embedding receptor conditioning fails a wrong-pocket control. The natural escape route — explicit 3D pocket geometry from crystal structures — is tested here with six pre-registered model arms, two evaluation protocols, and 280 trained models on 41,271 measured ChEMBL affinity pairs across eleven receptors, in 2.34 hours of CPU-only compute.

The result is a decisive, controlled negative: the 3D-grounded arm does not significantly outperform the wrong-pocket-3D control on the scaffold-disjoint protocol ($+0.00025$, $p = 0.221$) and is significantly worse on the temporal protocol (-0.0090, $p = 0.041$). Two practical positives emerge: AlphaFold DB predicted structures are interchangeable with crystal structures as geometry sources (pooled $\Delta r = -0.003$), and the null is stable across pocket radii (5.0–8.0 Å). A variance decomposition shows receptor identity explains 84.5% of outcome variance versus 2.7% for model arm — a direct, actionable message for practitioners.

We believe this fits JCIM's scope and its demonstrated interest in honest, controlled evaluations of machine learning for chemical information: the study is fully pre-registered, contains zero simulated data, reports a not-run probe transparently, maintains a public defect register, and releases every frozen artefact with SHA-256 checksums so that any reader can check, extend, or overturn the conclusion.

The manuscript is not under consideration elsewhere. The preprint is archived at `https://doi.org/10.6084/m9.figshare.35010655` and cited in the Data and Software Availability section.

Sincerely,

Md Mushfiqur Rahim
Independent Researcher, Bogura, Bangladesh
ORCID 0009-0003-1021-685X
`20mahin2020@gmail.com`